Journal Pre-proof



Number of Children and Nine-Year Health Trajectories Among Older Adults in China: The Role of Social Participation

Alex Pak Lik Tsang , Ka Yan Lai , Yue Hu

PII: S2667-0321(26)00033-8
DOI: https://doi.org/10.1016/j.ahr.2026.100304
Reference: AHR 100304

To appear in: *Aging and Health Research*

Received date: 15 July 2026
Revised date: 2 November 2026
Accepted date: 4 November 2026

Please cite this article as: Alex Pak Lik Tsang , Ka Yan Lai , Yue Hu , Number of Children and Nine-Year Health Trajectories Among Older Adults in China: The Role of Social Participation, *Aging and Health Research* (2026), doi: https://doi.org/10.1016/j.ahr.2026.100304

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2026 Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license
(http://creativecommons.org/licenses/by-nc-nd/4.0/)

Journal Pre-proof

Highlights

- Five survey waves tracked older Chinese parents over nine years.
- More children were associated with poorer self-rated and functional health.
- More children were associated with faster functional decline.
- Social participation's moderation of self-rated health was model-sensitive.
- No moderation was found for functional health or change in either health outcome.

Number of Children and Nine-Year Health Trajectories Among Older Adults in China: The Role of Social Participation

Alex Pak Lik Tsang, Ph.D.[1], Ka Yan Lai, BSc.[2], Yue Hu, Ph.D.[3,*]

[1]Department of Sociology and Social Policy, Lingnan University

[2]Department of Social and Behavioral Sciences, City University of Hong Kong

[3]The Nethersole School of Nursing, The Chinese University of Hong Kong

Alex Pak Lik Tsang: https://orcid.org/0000-0003-2919-1587

Ka Yan Lai: https://orcid.org/0009-0009-8554-7534

Yue Hu: https://orcid.org/0000-0002-8791-4512

[*] Corresponding author: Yue Hu

The Nethersole School of Nursing, The Chinese University of Hong Kong

Sha Tin, Hong Kong

Email: yuehu@cuhk.edu.hk

Journal Pre-proof

Journal Pre-proof

Keywords: number of children, self-rated health, functional limitations, social participation, older adults, longitudinal

Acknowledgments

The authors would like to thank the participants of CHARLS.

Journal Pre-proof

Journal Pre-proof

Statements and Declarations

Consent to Participate

CHARLS respondents provided written informed consent. The present study used publicly available, de-identified data.

Consent for Publication

Not applicable.

Ethical Approval

CHARLS was approved by the Biomedical Ethics Review Committee of Peking University (IRB00001052-11015), and all respondents provided written informed consent. The present study is a secondary analysis of publicly available, de-identified data.

Data Availability Statement

The CHARLS data are publicly available to registered users at https://charls.pku.edu.cn/en.

Journal Pre-proof

Number of Children and Nine-Year Health Trajectories Among Older Adults in China: The Role of Social Participation

Journal Pre-proof

Abstract

Background Traditional Chinese norms favor larger families, yet their implications for later-life health remain unclear. We examined associations of the number of children with self-rated health (SRH), functional health, and changes in both outcomes, and moderation by social participation.

Methods We used five waves (2011–2020) of the China Health and Retirement Longitudinal Study. The regression sample comprised 5,788 respondents aged ≥60 years with at least one living child. Social participation was assessed at baseline and dichotomized as any participation versus no participation in social activities. Mixed-effects ordinal logistic models tested number of children × social participation and × time interactions.

Results More children were associated with poorer SRH (OR = 1.080, 95% CI [1.030, 1.130]), more instrumental activities of daily living (IADL) limitations (OR = 1.070, 95% CI [1.020, 1.130]), and faster accumulation of IADL limitations (OR = 1.013 per year, 95% CI [1.007, 1.020]), but not faster SRH decline. Social participation was associated with better SRH (OR = 0.740, 95% CI [0.670, 0.810]) and fewer IADL limitations (OR = 0.630, 95% CI [0.560, 0.700]). The association between the number of children and SRH was weaker with social participation (interaction OR = 0.940, 95% CI [0.880, 1.000]), but was sensitive to model specification. Social participation did not significantly moderate the association with IADL limitations or the rate of change in either health outcome.

Conclusions Older adults with more children may be at risk of poorer SRH and faster functional decline. Whether promoting social participation reduces these disparities requires intervention studies.

Keywords: number of children, self-rated health, functional limitations, social participation, older adults, longitudinal

Journal Pre-proof

Journal Pre-proof

1. Introduction

In line with global demographic shifts, China is experiencing declining fertility and population aging [1]. By 2050, the population aged 65 years and older in China is projected to reach 395 million [2]. Traditional Chinese culture advocates “the more children, the greater prosperity”, and for Chinese older adults, having more children is often perceived as indicating a stronger support network. However, the demands of a large family may impose multiple and accumulating strains over a person’s life, which in turn may elevate the likelihood of poor health at a later stage. Emerging studies have examined the association between the number of children and health among older adults [3]. However, key questions remain regarding the long-term association between the number of children and older adults’ self-rated and functional health, and the conditions under which this association may be weakened.

The World Health Organization conceptualizes health as encompassing physical, mental, and social well-being [4]. This multidimensional perspective highlights the value of examining complementary health outcomes in later life. Self-rated health (SRH) captures individuals’ overall appraisal of their health [5], whereas functional health, commonly indexed by limitations in instrumental activities of daily living (IADL), reflects their ability to perform activities necessary for independent living [6]. Examining both outcomes therefore allows us to assess whether the number of children and its interaction with social participation are associated with older adults’ overall perceptions of health and their everyday functioning.

Evidence on whether a higher number of children is associated with poorer health among older adults is mixed. Having children has been associated with lower mortality compared with childlessness [7], but among parents, a higher number of children has been associated with a greater risk of cardiovascular disease [8-9] and hospitalization [7], and associations with cognitive function appear non-linear [10-11]. In China, cross-sectional studies have linked a higher number of children to poorer SRH among middle-aged and older women [12] and migrant women [13], and to more depressive

Journal Pre-proof

symptoms among rural older women [14]. Using CHARLS, an instrumental-variable study found that more children predicted poorer later-life health, although the association was sensitive to adjustment for education [15]. Conversely, rural older adults with more children have been found to have better functional health [16-17], associations attributed to intergenerational support. Longitudinal evidence is scarce: in CHARLS, women whose family trajectories combined early marriage, and three or more children showed a faster increase in physical functional limitations from midlife [18]. Most studies were cross-sectional or restricted to women or rural residents, and to our knowledge none has examined whether the number of children is associated with the rate of change in both SRH and functional health in the broader older population.

Two theoretical perspectives lead to competing expectations. From a support perspective, each additional child is a potential source of intergenerational support [7,16,17]. From a strain perspective, role strain theory [19] suggests that people experience role strain, defined as the perceived difficulty in meeting role obligations, when the demands of their roles exceed their available resources. Having multiple children increases parental stress [20], and older parents of several children may face diversified demands from them. Support may predominate when children are able and available to help, whereas strain may predominate when the demands of children, such as financial assistance or grandchild care, exceed parents' resources [19]. Given that most previous studies of SRH point to poorer health with more children [12-15], we expected the strain pathway to predominate in the cohorts studied here.

Social participation refers to a person's involvement in activities that provide interaction with others in society or the community [21]. It has been associated with lower mortality [22], fewer functional disabilities [23], and better SRH and physical function [24]. Beyond these direct benefits, social participation may also modify the health consequences of stress. The stress-buffering hypothesis

Journal Pre-proof

proposes that social relationships protect individuals from the adverse effects of stressful circumstances by providing emotional, informational, and instrumental resources [25]. Social participation fosters relationships that meet the basic psychological need of relatedness [26] and provides sources of support outside the family, which may reduce reliance on children and offset the demands that arise from them. We therefore proposed that social participation would weaken the association between having more children and poorer health. In a recent cross-sectional study of Chinese older adults, those with more children reported lower social participation, which partly accounted for their poorer cognitive function [11]. That study treated social participation as a pathway; to our knowledge, no study has examined whether social participation modifies the association between the number of children and SRH or functional health, or whether it does so over time.

This study examined whether the number of children was associated with SRH and functional health among older adults in China over nine years, and whether social participation modified these associations. Four hypotheses were tested for each outcome. H1a: A higher number of children would be associated with poorer SRH and more IADL limitations across waves. H1b: A higher number of children would be associated with a faster decline in SRH and a faster accumulation of IADL limitations. H2a: Social participation would weaken the associations between the number of children and each health outcome. H2b: Social participation would weaken the associations between the number of children and the rate of change in each outcome. Given previous reports of non-linear associations with cognitive function [10,11,27], we also explored whether the association was non-linear.

2. Methods

2.1 Study Design and Participants

This prospective cohort study used data from the China Health and Retirement Longitudinal Study (CHARLS), a nationally representative survey of adults aged 45 years and older in 28 provinces, selected by multistage probability sampling [28]. The baseline survey (T1) was conducted in 2011,

Journal Pre-proof

followed by waves in 2013 (T2), 2015 (T3), 2018 (T4), and 2020 (T5). Of the 25,586 baseline respondents, 16,730 were excluded as aged below 60 years or with missing age, 199 as childless, and 1,317 for a missing child count. Childless respondents were excluded because they were not exposed to the hypothesized mechanism of role strain and including them as a zero category would assume that the difference between no children and one child is equivalent to the differences between adjacent counts among parents; their inclusion was examined in a sensitivity analysis. To estimate within-person changes, respondents were required to report valid data on both SRH and IADL limitations at two or more waves; 847 with fewer than two valid waves on at least one outcome were excluded. The final analytical sample comprised 6,493 respondents (Figure 1), of whom 5,788 had complete data on social participation and on all covariates other than household income and contributed 23,893 SRH and 24,943 IADL observations to the regression models.

2.2 Measures

2.2.1 Health

SRH was assessed with a single item ("Would you say your health is very good, good, fair, poor, or very poor?") scored from 1 (very good) to 5 (very poor). Functional health was measured with five items adapted from the Lawton IADL Scale [6] including household chores, meal preparation, shopping, taking medications, and managing money. The telephone item was unavailable at T1 and was excluded from all waves. Items were rated on a four-point scale and dichotomized as any difficulty versus none, and the number of items with difficulty was summed (range 0–5) and modeled on its full range. Cronbach's alpha ranged from 0.79 to 0.84 across waves.

2.2.2 Number of Children

The number of children was the total number of living children reported at T1, derived from the CHARLS family roster, in which each child is enumerated with sex and survival status. The variable is recorded at the household level: both partners in a couple are assigned the same value, and the count includes all living children irrespective of co-residence. The roster does not distinguish biological,

Journal Pre-proof

adopted, and stepchildren, and the measure does not capture the duration, timing, or intensity of child caregiving. Because the count excludes deceased children, parity was examined in a sensitivity analysis (r = 0.95 with the number of living children).

2.2.3 Social Participation

Respondents reported on whether they had engaged in six activities in the past month at T1: interacting with friends; playing Mahjong, chess, or cards; attending community or sports clubs; participating in community organizations; volunteer or charity work; and educational or training courses. Participation in any activity was coded 1 and none 0. The binary indicator was retained as the primary measure for consistency with previous CHARLS research [29] and because 88.00% of respondents reported no more than one activity (Table 1). Baseline participation was used because participation at later waves is plausibly a consequence of intervening health change [29]. A count of activities (0–6) and time-varying participation were examined in sensitivity analyses. A frequency-weighted measure was not constructed because frequency was recorded only among participants (Supplementary Methods S1).

2.2.4 Covariates

Covariates were selected on the basis of previous studies of SRH and functional limitations [30] and were measured at T1: age, gender, education (below lower secondary, upper secondary or vocational, or tertiary), marital status (six categories), household registration (urban or rural), annual household income (five categories), pension (yes or no), alcohol consumption frequency (five categories), smoking status (never, former, or current), and chronic illness, coded as any versus none of eleven conditions (Table 1). Because these covariates were measured after family size was determined and may partly have been shaped by it, estimates are interpreted as associations conditional on baseline socioeconomic and health status rather than as total effects.

Journal Pre-proof

Journal Pre-proof

Three sets of variables were not adjusted for in the primary models. First, depressive symptoms, co-residence with children, and financial transfers from children plausibly lie on the pathway from the number of children to health or are consequences of health, so adjusting for them would risk overadjustment [31]. The sex composition of children was also not adjusted for because it is determined jointly with the number of children. Second, childhood socioeconomic circumstances were available only from the 2014 Life History Survey and were missing for 11.00% of the sample. Both sets were examined in sensitivity analyses. Third, baseline SRH and IADL were not entered as covariates because the baseline wave is modeled as an outcome observation; between-person differences in baseline health and in the rate of change are captured by the random intercept and random slope.

2.3 Data Analyses

Analyses were performed in Stata 18 [32]. Mixed-effects ordinal logistic regression was used because both outcomes are ordinal (the IADL score is a bounded count with many zeros, for which a linear model is not tenable) and repeated observations are nested within respondents. Time was coded as years since baseline (0, 2, 4, 7, and 9) and entered as linear and quadratic terms, the latter constructed as the square of the uncentered linear term. The number of children was mean centered (M = 3.36, SD = 1.52). Models were estimated by maximum likelihood with mean–variance adaptive Gauss–Hermite quadrature (seven integration points), which uses all available outcome observations under the missing-at-random assumption. Household income was missing for 26.00% of the sample; missingness was retained as an explicit category (the missing-indicator method), and respondents with missing values on social participation or other covariates were excluded.

Three specifications were fitted in sequence. Model 1 included the number of children, social participation, their interaction, linear and quadratic time, and covariates, with a random intercept. Model 2 added a number of children × time interaction, which tests whether the number of children predicts differential change. Model 3 added the three-way interaction among number of children, social

Journal Pre-proof

participation, and time, which tests whether social participation moderates health trajectories. Likelihood-ratio tests indicated that a random slope for time was required, so Models 2 and 3 included a random slope with an unstructured covariance; to achieve convergence, time was standardized (*SD* = 3.26 years) in these models, and time interactions are reported per year. Results are reported as odds ratios (ORs) with 95% confidence intervals (CIs) and predicted probabilities and average marginal effects were computed to aid interpretation. Sampling weights were not applied because the estimand is a conditional association and CHARLS does not release the level-specific weights required for weighted multilevel models (Supplementary Methods S2).

2.4 Sensitivity Analyses

The proportional-odds assumption was assessed with cluster-robust Wald tests, and threshold-specific models imposing no proportionality constraint were estimated. Non-linearity in the exposure was examined with a quadratic term and a categorical specification with one child as the reference. Further analyses used (a) parity in place of living children; (b) a count of activities; (c) time-varying participation; (d) a three-level model with community-level random intercepts; (e) childless respondents as an additional exposure category; (f) sex-stratified models; (g) additional adjustment for depressive symptoms, co-residence, financial transfers, and the sex composition of children; (h) adjustment for childhood socioeconomic circumstances; (i) restriction to waves 1 to 4; and (j) a comparison of respondents with and without observed income, with stratified models and a comparison of the missing-indicator method, complete-case analysis, and multiple imputation. Details are given in Supplementary Methods S3.

3. Results

3.1 Participant Characteristics

Table 1 presents the baseline characteristics of the 6,493 respondents. The mean age was 67.71 years (*SD* = 6.44), 50.50% were women, 93.50% had less than lower secondary education, 76.80% were

Journal Pre-proof

married, 62.20% lived in rural villages, and 69.70% reported at least one chronic condition. The number of children ranged from 1 to 10 (median = 3.00; *M* = 3.36, *SD* = 1.52). At baseline, 54.70% reported no social activity in the past month; participation was concentrated in interacting with friends (33.80%) and playing Mahjong, chess, or cards (16.50%), and the mean number of activities was 0.59 (*SD* = 0.74). Poor or very poor SRH was reported by 31.70% at T1 and 31.80% at T5, and the proportion without IADL limitations declined from 73.00% at T1 to 64.10% at T5. Missing data ranged from 9.90% for the outcomes at T2 to 26.00% for household income.

3.2 Self-Rated Health

In Model 1 (Table 2), a greater number of children was associated with poorer SRH (OR = 1.080, 95% CI [1.030, 1.130], $p = 0.001$), and social participation with better SRH (OR = 0.740, 95% CI [0.670, 0.810], $p < 0.001$). The number of children × social participation interaction was negative (OR = 0.940, 95% CI [0.880, 1.000], $p = 0.048$). Among respondents without social participation, the predicted probability of poor SRH rose from 0.250 (95% CI [0.230, 0.260]) at one child to 0.280 ([0.270, 0.300]) at six children, whereas among participants it changed little (0.230 to 0.240; Figure 2A; very poor SRH, Figure S2). The average marginal effect of one additional child on the probability of poor SRH was 0.008 (95% CI [0.003, 0.012]) without social participation and 0.001 (95% CI [−0.003, 0.006]) with social participation, a difference of −0.006 (95% CI [−0.012, −0.000], $p = 0.049$; Figure 2B–C). The interaction did not remain significant with a random slope for time ($p = 0.054$) or with community-level clustering ($p = 0.082$; Table S4), so it is consistent in direction but imprecisely estimated.

SRH improved slightly over approximately the first three years of follow-up and worsened at an accelerating rate thereafter (Table 2). The number of children × time interaction was not significant (OR = 1.006 per year, 95% CI [0.999, 1.012], $p = 0.095$), nor was the three-way interaction ($\chi^2(1) = 0.65$, $p = 0.420$; Table S1). The two-way interaction therefore reflected effect modification of the average association across waves.

Journal Pre-proof

3.3 IADL Limitations

A greater number of children was associated with more IADL limitations (OR = 1.070, 95% CI [1.020, 1.130], $p = 0.011$), and social participation with fewer (OR = 0.630, 95% CI [0.560, 0.700], $p < 0.001$; Table 2). The number of children × social participation interaction was not significant (OR = 0.970, 95% CI [0.900, 1.040], $p = 0.384$). The number of children × time interaction was positive (OR = 1.013 per year, 95% CI [1.007, 1.020], $p < 0.001$; random-slope model, OR = 1.020, 95% CI [1.011, 1.028]; Table S1): the predicted probability of no IADL limitation declined from approximately 0.730 to 0.630 over nine years among respondents with one child, and from 0.730 to 0.550 among those with five (Figure 3; five limitations, Figure S3). This pattern held in waves 1 to 4 (OR = 1.022 per year, 95% CI [1.013, 1.030]). The three-way interaction was not significant ($\chi^2(1) = 1.08$, $p = 0.300$).

3.4 Sensitivity Analyses

The proportional-odds assumption was violated mainly by the time and age terms, and not by the number of children (Table S2). In threshold-specific models, the SRH interaction was concentrated at the upper end of the health distribution (OR = 0.860, 95% CI [0.770, 0.970] for less than very good health; OR = 0.990, 95% CI [0.920, 1.080] for poor or worse health; Table S3; Figure S4), and no IADL interaction was significant. A quadratic term in the number of children was not significant for either outcome among non-participants, but for SRH it differed by participation (OR = 1.035, 95% CI [1.006, 1.066], $p = 0.020$). In the categorical specification, the odds of poorer SRH among non-participants increased progressively with the number of children (OR = 1.000, 1.060, 1.200, and 1.390 for two, three, four, and five or more children), whereas among participants the pattern was flatter and influenced by the one-child group (Figure S1; Supplementary Results S1).

The SRH interaction was comparable with parity in place of living children (OR = 0.940, 95% CI [0.890, 1.000]), with participation measured as a count (OR = 0.940 per activity, 95% CI [0.900, 0.990], $p = 0.009$) or as time-varying (OR = 0.960, 95% CI [0.920, 1.000]), with adjustment for childhood

Journal Pre-proof

socioeconomic circumstances (OR = 0.920, 95% CI [0.860, 0.980]), with childless respondents included, and in waves 1 to 4 (OR = 0.940, 95% CI [0.880, 1.010]; Table S4). Additional adjustment for post-exposure variables attenuated it (OR = 0.960, 95% CI [0.900, 1.010], p = 0.130). The estimates were 0.890 (95% CI [0.810, 0.990]) among men and 0.960 (95% CI [0.890, 1.050]) among women; the difference was not formally tested. Single-level missing-indicator and multiple-imputation estimates were similar; complete-case analysis used fewer observations (Table S4). The interaction differed between respondents with observed and missing income (OR = 0.900 and 1.120; Supplementary Results S2).

4. Discussion

Using five waves of CHARLS data spanning nine years, this study found that a higher number of children was associated with poorer SRH and more IADL limitations across waves, supporting H1a, and with faster accumulation of IADL limitations, supporting H1b for functional health but not for SRH. Social participation was associated with better health on both outcomes. The association between the number of children and SRH was weaker among those reporting social participation, partially supporting H2a, but this interaction was imprecisely estimated. Social participation did not modify the association with IADL limitations or the rate of change in either outcome, so H2b was not supported. Because the design is observational, these findings describe associations rather than effects.

The association between more children and poorer parental health is consistent with previous studies of SRH in China [12-14] and with cohort evidence linking a larger number of children to adiposity and coronary heart disease in both mothers and fathers [8] and to maternal cardiovascular disease [9]. Among non-participants, the odds of poorer SRH rose progressively with the number of children, and the U-shaped pattern reported for cognitive function [10,11,27] was not observed. The present study extends this evidence by showing that the gap in functional health between parents with fewer and more children widened over nine years, consistent with the faster increase in functional limitations

Journal Pre-proof

among Chinese women whose family trajectories combined early marriage and three or more children [18], although not all studies agree on the direction of the association [16-17].

One possible explanation is role strain arising from the diversified demands of children. Some older adults are expected to provide grandchild care, sometimes migrating or living apart from their partners to do so [33-34], and the associated strain may threaten their health [34]. Because grandchild caregiving and role strain were not measured, this account remains speculative. Social participation provides opportunities to connect with others emotionally, behaviorally, and physically [35], which may help older parents cope with these demands; this is consistent with the weaker SRH association among participants, although the design cannot establish buffering. That social participation modified the association with SRH but not with IADL limitations may reflect differences in sensitivity: SRH integrates diverse health information within each individual's frame of reference [5] and may respond to the positive mood associated with social contact, whereas functional limitations depend on chronic disease and cumulative physical and cognitive capacity [36]. The null IADL interaction may also reflect the binary measure of participation or limited power to detect interactions.

Alternative explanations should also be considered. First, the number of children may partly reflect fertility selection: early-adult health, socioeconomic circumstances, and preferences may influence both family size and later-life health. In England, associations between parity and later-life cognition were largely accounted for by socioeconomic position [27], and in CHARLS the association between the number of children and health was sensitive to adjustment for education [15]. Adjustment for baseline socioeconomic status and, in a subsample, childhood circumstances did not alter the present findings, but earlier-life circumstances were only partly observed. Second, these cohorts formed their families before or during the early years of China's family planning policies, and the meaning and demands of having several children may differ across cohorts. Third, adjustment for co-residence,

Journal Pre-proof

financial transfers, and other post-exposure variables attenuated the interaction, consistent with these variables partly accounting for it.

This study used a large, nationally representative sample followed over nine years, examined two complementary outcomes, tested differences in the rate of change as well as the level of health, and examined the robustness of the findings across a range of sensitivity analyses.

Several limitations should be noted. First, the number of living children does not capture stress, role strain, the demands of adult children or grandchild care, or the timing and intensity of child caregiving, and features of fertility history such as age at first birth have also been linked to later-life health [27,37]. Because the roster does not distinguish biological, adopted, and step-children, biological and social pathways cannot be separated [8]. Future studies should measure stress and stressful family events directly. Second, the number of children and social participation were measured at baseline, and the binary participation measure disregards the type, frequency, and intensity of activities; a count and a time-varying specification yielded comparable estimates. Participation was nearly unrelated to the number of children ($r = -0.02$), so the interaction is unlikely to reflect conditioning on a mediator. Third, all measures were self-reported, and IADL limitations were not assessed with performance-based tests. Fourth, household income was missing for 26.0% of respondents. The missing-indicator method can yield biased estimates [38], but it agreed closely with multiple imputation; nonetheless, the moderation estimate diverged between respondents with and without observed income, which adds uncertainty. Fifth, respondents lost by T5 were older, in poorer health, and had more children than those retained (Supplementary Results S3), so healthy survivor bias may have led to underestimation of the association with functional decline. Sixth, the sex-stratified estimates are descriptive; a significant estimate in one group alongside a non-significant estimate in the other does not show that they differ [39]. Seventh, the exclusion of childless adults limits the findings to older Chinese parents, and because the study is

Journal Pre-proof

Journal Pre-proof

observational, residual confounding may remain. Finally, the findings may not apply to younger cohorts: in CHARLS, respondents aged 75 years and older had nearly four living children on average, compared with about two among those aged 55–60 [2].

The findings highlight the importance of attending to the health needs of older adults with larger families. Efforts to support these older adults could consider opportunities for social participation alongside support for maintaining independence in daily living. The potential role of social participation in addressing health disparities associated with having more children warrants further investigation, particularly because the observed moderation was sensitive to model specification. Intervention studies could assess whether increasing participation improves self-rated and functional health in this population and whether any benefits differ according to the number of children.

5. Conclusions

Using five-wave longitudinal data spanning nine years and covering 6,493 respondents aged 60 years and older, with 5,788 respondents in the primary regression models, we found that a higher number of children was associated with worse self-rated health and more IADL limitations, after controlling for baseline sociodemographic, economic, and health-related covariates. Respondents with more children also accumulated functional limitations faster over time. Social participation was associated with better self-rated health and fewer IADL limitations, and the association between the number of children and self-rated health was weaker among those who participated, although this moderation was imprecisely estimated and sensitive to model specification. Social participation did not modify the association with functional health or the rate of change in either outcome.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Journal Pre-proof

Journal Pre-proof

Ethics Statement

CHARLS was approved by the Biomedical Ethics Review Committee of Peking University (IRB00001052-11015), and all respondents provided written informed consent [28]. The present study is a secondary analysis of publicly available, de-identified data.

Data Availability

The CHARLS data are publicly available to registered users at https://charls.pku.edu.cn/en.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Acknowledgments

The authors would like to thank the participants of CHARLS.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author(s) used ChatGPT for language editing. The author(s) reviewed and edited the output as needed and take full responsibility for the content of the published article.

Journal Pre-proof

Journal Pre-proof

References

[1] Li L, Jin G, Lai X, Jing R, Zhu H. A reassessment of trends and rural–urban/regional differences in the total fertility rate in China, 2000–2020: analyses of the 2020 national census data. Sci Rep 2024;14(1):8601. https://doi.org/10.1038/s41598-024-59177-2.

[2] Chen X, Giles J, Yao Y, Yip W, Meng Q, Berkman L, et al. The path to healthy ageing in China: a Peking University–Lancet Commission. Lancet 2022;400(10367):1967-2006. https://doi.org/10.1016/S0140-6736(22)01546-X.

[3] van den Broek T, Fleischmann M. The causal effect of number of children on later-life overweight and obesity in parous women. An instrumental variable study. Prev Med Rep 2021;24:101528. https://doi.org/10.1016/j.pmedr.2021.101528.

[4] World Health Organization. Constitution of the World Health Organization. Am J Public Health Nations Health 1946;36(11):1315-23. https://doi.org/10.2105/AJPH.36.11.1315.

[5] Jylhä M. What is self-rated health and why does it predict mortality? Towards a unified conceptual model. Soc Sci Med 2009;69(3):307-16. https://doi.org/10.1016/j.socscimed.2009.05.013.

[6] Lawton MP, Brody EM. Assessment of older people: self-maintaining and instrumental activities of daily living. Gerontologist 1969;9(3 Part 1):179-86. https://doi.org/10.1093/geront/9.3_Part_1.179.

[7] Meyer AC, Brooke HL, Modig K. The role of children and their socioeconomic resources for the risk of hospitalisation and mortality – a nationwide register-based study of the total Swedish population over the age 70. BMC Geriatr 2019;19(1):114. https://doi.org/10.1186/s12877-019-1134-y.

Journal Pre-proof

Journal Pre-proof

[8] Lawlor DA, Emberson JR, Ebrahim S, Whincup PH, Wannamethee SG, Walker M, et al. Is the association between parity and coronary heart disease due to biological effects of pregnancy or adverse lifestyle risk factors associated with child-rearing?: findings from the British Women's Heart and Health Study and the British Regional Heart Study. Circulation 2003;107(9):1260-4. https://doi.org/10.1161/01.CIR.0000053441.43495.1A.

[9] Li W, Ruan W, Lu Z, Wang D. Parity and risk of maternal cardiovascular disease: a dose–response meta-analysis of cohort studies. Eur J Prev Cardiol 2019;26(6):592-602. https://doi.org/10.1177/2047487318818265.

[10] Saenz JL, Díaz-Venegas C, Crimmins EM. Fertility history and cognitive function in late life: the case of Mexico. J Gerontol B Psychol Sci Soc Sci 2021;76(4):e140-52. https://doi.org/10.1093/geronb/gbz129.

[11] Yu Z, Jiawen P, Jia M, Kaixuan X, Youpeng G, Ting L, et al. Gender differences in the mediating effects of social participation and number of children on psychological resilience and cognitive function in older adults. Geriatr Nurs 2025;63:186-92. https://doi.org/10.1016/j.gerinurse.2025.03.014.

[12] Li X, Jiang Q, Li S, Feldman MW. Female fertility history and mid-late-life health: findings from China. J Women Aging 2018;30(1):62-74. https://doi.org/10.1080/08952841.2016.1259445.

[13] Yang X, Xie L. Fertility and self-rated health of migrant women of childbearing age—an analysis of moderating effects based on socioeconomic status. BMC Womens Health 2024;24(1):238. https://doi.org/10.1186/s12905-024-03043-w.

Journal Pre-proof

Journal Pre-proof

[14] Zhang S, Yang H, Li Z, Zhang S, Wu Y. A study of the effect of number of children on depression among rural older women: empirical evidence from China. J Biosoc Sci 2024;56(1):182-206. https://doi.org/10.1017/S0021932023000172.

[15] Weng Y, Yang X. Fertility behaviors and mid-late-life health status in China: from a life-course perspective. Soc Sci Med 2023;338:116314. https://doi.org/10.1016/j.socscimed.2023.116314.

[16] Yu C, Liang H, Wang B, Liang F, Liu E, Xiang N. The association between reproductive history and the multidimensional health of older adults in rural China and its gender differences: evidence from the Chinese longitudinal healthy longevity survey. Front Public Health 2022;10:952671. https://doi.org/10.3389/fpubh.2022.952671.

[17] Long C, Han J, Yi C. The health effect of the number of children on Chinese elders: an analysis based on hukou category. Front Public Health 2021;9:700024. https://doi.org/10.3389/fpubh.2021.700024.

[18] Chen CC, Zhou N, Wu XW, Qiao Y, Wang XB. Family life-course trajectories and longitudinal changes in depressive symptoms and physical functional limitations in Chinese women from midlife to early old age: a 7-year longitudinal cohort study. BMC Public Health 2026;26:2525. https://doi.org/10.1186/s12889-026-28482-y.

[19] Goode WJ. A theory of role strain. Am Sociol Rev 1960;25(4):483-96. https://doi.org/10.2307/2092933.

[20] Zhou H, Wang X, Li M. Parental stress, coercive and encouraging parenting among Chinese one-child, two-child and three-child families. Fam Relatsh Soc 2024;13(4):528-48. https://doi.org/10.1332/20467435Y2024D000000018.

Journal Pre-proof

Journal Pre-proof

[21] Levasseur M, Richard L, Gauvin L, Raymond É. Inventory and analysis of definitions of social participation found in the aging literature: proposed taxonomy of social activities. Soc Sci Med 2010;71(12):2141-9. https://doi.org/10.1016/j.socscimed.2010.09.041.

[22] Minagawa Y, Saito Y. Active social participation and mortality risk among older people in Japan: results from a nationally representative sample. Res Aging 2015;37(5):481-99. https://doi.org/10.1177/0164027514545238.

[23] Ashida T, Kondo N, Kondo K. Social participation and the onset of functional disability by socioeconomic status and activity type: the JAGES cohort study. Prev Med 2016;89:121-8. https://doi.org/10.1016/j.ypmed.2016.05.006.

[24] Zhou J, Yang Y, Li S, Mao N, Chen X, Wang D, et al. Patterns of social participation among older adults and their association with self-rated health: the mediating role of activities of daily living. Aging Clin Exp Res 2025;37(1):221. https://doi.org/10.1007/s40520-025-03131-3.

[25] Cohen S, Wills TA. Stress, social support, and the buffering hypothesis. Psychol Bull 1985;98(2):310-57. https://doi.org/10.1037/0033-2909.98.2.310.

[26] Ryan RM, Deci EL. Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. Am Psychol 2000;55(1):68-78. https://doi.org/10.1037/0003-066X.55.1.68.

[27] Read SL, Grundy EMD. Fertility history and cognition in later life. J Gerontol B Psychol Sci Soc Sci 2017;72(6):1021-31. https://doi.org/10.1093/geronb/gbw013.

[28] Zhao Y, Hu Y, Smith JP, Strauss J, Yang G. Cohort profile: the China Health and Retirement Longitudinal Study (CHARLS). Int J Epidemiol 2014;43(1):61-8. https://doi.org/10.1093/ije/dys203.

Journal Pre-proof

Journal Pre-proof

[29] Tsang APL, Lee CK, Chan SCY. A cross-lagged panel analysis of social participation in the relationship between functional limitations and cognitive functioning: evidence from CHARLS. J Appl Gerontol 2025;44(4):551–60. https://doi.org/10.1177/07334648241282756.

[30] Nóbrega JCL, Medeiros JB, Santos TTDM, Alves SAV, Freitas JLGDS, Silva JMM, et al. Socioeconomic factors and health status disparities associated with difficulty in ADLs and IADLs among long-lived populations in Brazil: a cross-sectional study. Inquiry 2021;58:00469580211007264. https://doi.org/10.1177/00469580211007264.

[31] Schisterman EF, Cole SR, Platt RW. Overadjustment bias and unnecessary adjustment in epidemiologic studies. Epidemiology 2009;20(4):488-95. https://doi.org/10.1097/EDE.0b013e3181a819a1.

[32] StataCorp. Stata 18 Base Reference Manual. College Station, TX: Stata Press; 2023.

[33] Hu Y, Wangliu Y, Fung HH, Bookwala J, Wong J, Tong Y. The profile of stress and coping related to successful aging among Chinese migrant and nonmigrant grandparents. Gerontologist 2024;64(4):gnad125. https://doi.org/10.1093/geront/gnad125.

[34] Hu Y, Kang R, Charness N, Chan ACY, Zhou S, You J, et al. Is Internal Migration Bad for Grandparents' Subjective Successful Aging? Activities, Adaptation & Aging. 2025 Oct 2;49(4):651–70. doi:10.1080/01924788.2025.2492979

[35] Holt-Lunstad J. Why social relationships are important for physical health: a systems approach to understanding and modifying risk and protection. Annu Rev Psychol 2018;69(1):437-58. https://doi.org/10.1146/annurev-psych-122216-011902.

Journal Pre-proof

Journal Pre-proof

[36] Stuck AE, Walthert JM, Nikolaus T, Büla CJ, Hohmann C, Beck JC. Risk factors for functional status decline in community-living elderly people: a systematic literature review. Soc Sci Med 1999;48(4):445-69. https://doi.org/10.1016/S0277-9536(98)00370-0.

[37] Pirkle CM, de Albuquerque Sousa ACP, Alvarado B, Zunzunegui MV; IMIAS Research Group. Early maternal age at first birth is associated with chronic diseases and poor physical performance in older age: cross-sectional analysis from the International Mobility in Aging Study. BMC Public Health 2014;14(1):293. https://doi.org/10.1186/1471-2458-14-293.

[38] Greenland S, Finkle WD. A critical look at methods for handling missing covariates in epidemiologic regression analyses. Am J Epidemiol 1995;142(12):1255-64. https://doi.org/10.1093/oxfordjournals.aje.a117592.

[39] Altman DG, Bland JM. Interaction revisited: the difference between two estimates. BMJ 2003;326(7382):219. https://doi.org/10.1136/bmj.326.7382.219.

Journal Pre-proof

Journal Pre-proof

Figure Captions

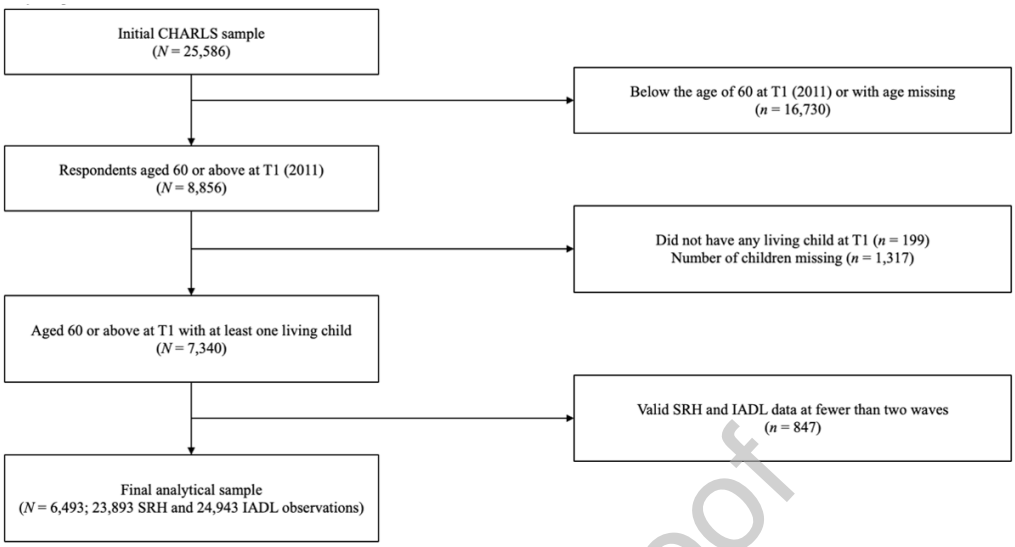


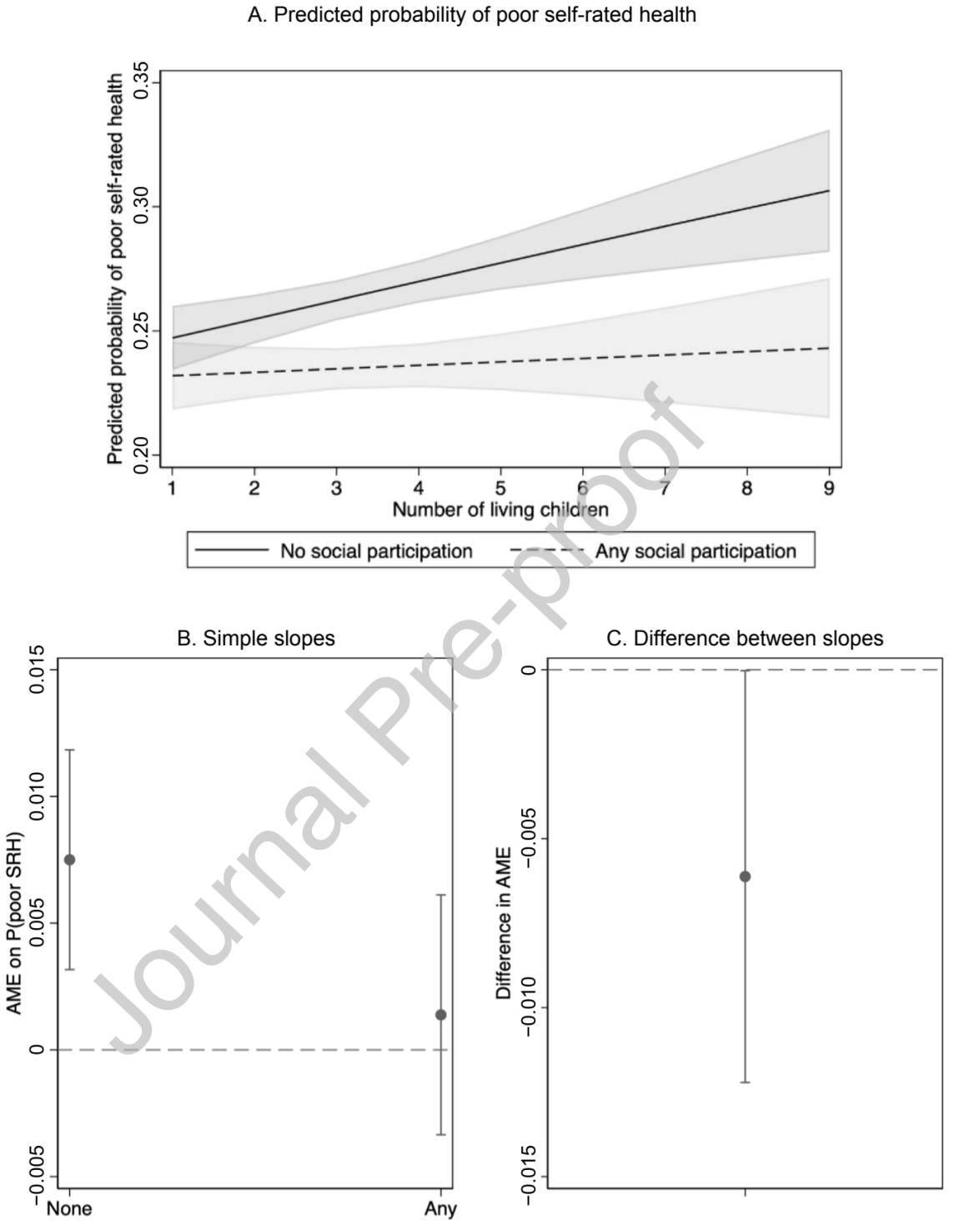
Figure 1

Flowchart of Sample Selection in CHARLS

Note. CHARLS = China Health and Retirement Longitudinal Study; SRH = self-rated health; IADL = instrumental activities of daily living.

Journal Pre-proof

Journal Pre-proof



Journal Pre-proof

Figure 2

Journal Pre-proof

Predicted Probability of Poor Self-Rated Health and Average Marginal Effect of the Number of Children, by Social Participation

Note. Panel A: population-averaged predicted probabilities of poor self-rated health from Model 1 (Table 2), with covariates at observed values; solid line = no social participation; dashed line = any social participation; shaded bands are 95% confidence intervals; estimates above six children are based on fewer than 250 respondents. Panel B: average marginal effect of one additional child on the probability of poor self-rated health within each social participation group. Panel C: difference between the two effects, which corresponds to the number of children × social participation interaction. Vertical bars are 95% confidence intervals, and dashed lines mark zero.

Journal Pre-proof

Journal Pre-proof

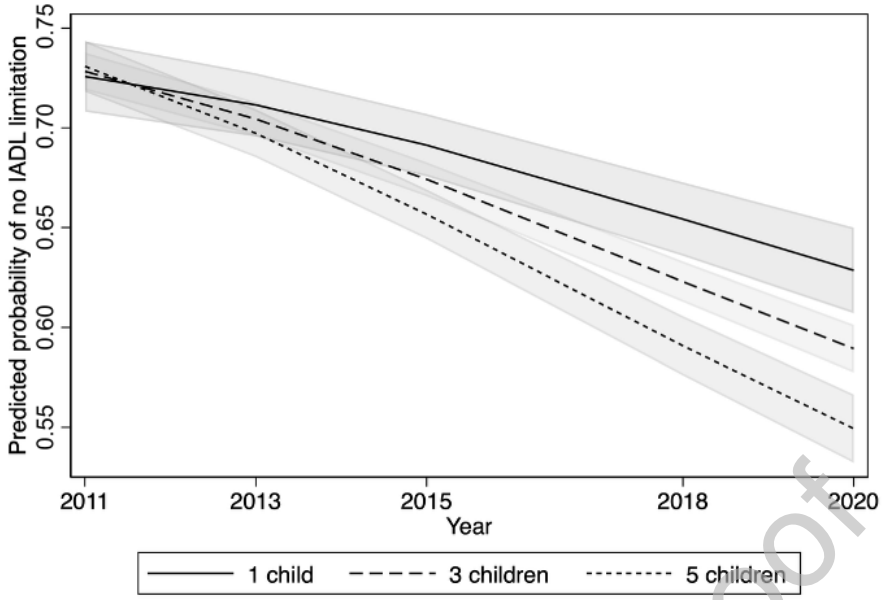


Figure 3

Predicted Probability of Reporting No IADL Limitation by Number of Children, 2011 to 2020

Note. Population-averaged predicted probabilities from the model including a number of children × time interaction and a random slope for time, adjusted for the covariates listed in Table 2. Shaded bands are 95% confidence intervals. IADL = instrumental activities of daily living.

Journal Pre-proof

Journal Pre-proof

Table 1

Baseline Socio-Demographic and Health Characteristics (N = 6,493)

Characteristic	
Age, *M* (*SD*)	67.71 (6.44)
Gender, n(%)	
Male	3,213 (49.48)
Female	3,280 (50.52)
Education level, n(%)	
Less than lower secondary	6,072 (93.52)
Upper secondary/vocational training	299 (4.60)
Tertiary	122 (1.88)
Marital status, n(%)	
Married	4,983 (76.78)
Married, spouse absent	244 (3.76)
Separated	44 (0.68)
Divorced	37 (0.57)
Widowed	1,175 (18.10)
Never married	7 (0.11)
Household registration (hukou), n(%)	
Urban community	2,454 (37.79)
Rural village	4,039 (62.21)
Annual household income, n(%)	
≤¥9,999	2,436 (37.52)

Journal Pre-proof

Journal Pre-proof

Characteristic	
¥10,000–19,999	639 (9.84)
¥20,000–29,999	473 (7.28)
¥30,000–39,999	361 (5.56)
≥¥40,000	896 (13.80)
Missing	1,688 (26.00)
Pension status, n(%)	
No	3,637 (56.45)
Yes	2,806 (43.55)
Alcohol consumption, n(%)	
Never	4,519 (74.62)
≤1 per month	423 (6.98)
1–3 per month	166 (2.74)
1–6 per week	234 (3.86)
≥1 per day	714 (11.79)
Smoking status, n(%)	
Never	3,773 (59.54)
Former	692 (10.92)
Current	1,872 (29.54)
Chronic illness, n(%)	
No	1,958 (30.26)
Yes	4,512 (69.74)
Number of chronic conditions, *M* (*SD*)	1.32 (1.27)

Journal Pre-proof

Journal Pre-proof

Characteristic	
Number of living children, n(%)	
1	603 (9.29)
2	1,421 (21.89)
3	1,743 (26.84)
4	1,360 (20.95)
5	827 (12.74)
6	334 (5.14)
7	136 (2.09)
8	48 (0.74)
9	14 (0.22)
10	7 (0.11)
Social participation, n(%)	
Any participation	2,789 (45.28)
Interacted with friends	2,084 (33.83)
Played Mahjong, chess, or cards	1,016 (16.49)
Attended a sport or social club	413 (6.70)
Participated in a community organization	88 (1.43)
Volunteer or charity work	25 (0.41)
Attended an educational or training course	4 (0.06)
Number of social activities, n(%)	
0	3,371 (54.72)
1	2,052 (33.31)
2	644 (10.45)

Journal Pre-proof

Journal Pre-proof

Characteristic	
3 or more	93 (1.51)
Number of activities, *M* (*SD*)	0.59 (0.74)

Note. Percentages are calculated over non-missing cases. Missing values: marital status, 3; pension status, 50; alcohol consumption, 437; smoking status, 156; chronic illness, 23; social participation (binary), 333; number of social activities, 333. Activity categories are not mutually exclusive, so the individual rates sum to more than the proportion reporting any participation.

Journal Pre-proof

Journal Pre-proof

Table 2

Mixed-Effects Ordinal Logistic Regression of Self-Rated Health and IADL Limitations on Number of Children and Social Participation (Model 1; N = 5,788)

Predictor	SRH, OR [95% CI]	*p*	IADL, OR [95% CI]	*p*
Number of children	1.080 [1.030, 1.130]	0.001	1.070 [1.020, 1.130]	0.011
Participation in social activities				
No	ref.		ref.	
Yes	0.740 [0.670, 0.810]	< 0.001	0.630 [0.560, 0.700]	< 0.001
Number of children × social participation	0.940 [0.880, 1.000]	0.048	0.970 [0.900, 1.040]	0.384
Time (linear)	0.950 [0.920, 0.980]	< 0.001	1.170 [1.130, 1.210]	< 0.001
Time (quadratic)	1.009 [1.006, 1.012]	< 0.001	0.997 [0.993, 1.000]	0.089
Age	1.010 [1.000, 1.020]	0.141	1.110 [1.100, 1.120]	< 0.001
Gender				
Male	ref.		ref.	
Female	1.270 [1.110, 1.440]	< 0.001	1.890 [1.630, 2.210]	< 0.001
Education level				
Less than lower secondary	ref.		ref.	
Upper secondary/vocational training	0.860 [0.680, 1.090]	0.213	0.550 [0.410, 0.750]	< 0.001
Tertiary	0.710 [0.490, 1.020]	0.066	0.390 [0.240, 0.650]	< 0.001
Marital status				
Married	ref.		ref.	
Married, spouse absent	1.050 [0.770, 1.430]	0.769	0.940 [0.650, 1.370]	0.763

Journal Pre-proof

Journal Pre-proof

Predictor	SRH, OR [95% CI]	*p*	IADL, OR [95% CI]	*p*
Separated	1.910 [1.030, 3.530]	0.039	1.280 [0.640, 2.570]	0.480
Divorced	1.160 [0.610, 2.200]	0.650	0.850 [0.390, 1.860]	0.682
Widowed	0.820 [0.720, 0.940]	0.004	0.920 [0.790, 1.070]	0.302
Never married	3.140 [0.710, 13.820]	0.130	5.700 [1.150, 28.230]	0.033
Household registration (hukou)				
Urban community	ref.		ref.	
Rural village	1.390 [1.250, 1.550]	< 0.001	1.440 [1.270, 1.630]	< 0.001
Annual household income				
≤¥9,999	ref.		ref.	
¥10,000–19,999	0.910 [0.760, 1.080]	0.269	1.100 [0.910, 1.340]	0.335
¥20,000–29,999	0.830 [0.680, 1.010]	0.056	1.000 [0.790, 1.250]	0.968
¥30,000–39,999	0.830 [0.660, 1.030]	0.095	0.820 [0.630, 1.070]	0.142
≥¥40,000	0.610 [0.520, 0.720]	< 0.001	0.650 [0.530, 0.780]	< 0.001
Missing	0.760 [0.670, 0.870]	< 0.001	0.750 [0.650, 0.870]	< 0.001
Pension status				
No	ref.		ref.	
Yes	0.800 [0.720, 0.890]	< 0.001	0.750 [0.660, 0.850]	< 0.001
Alcohol consumption				
Never	ref.		ref.	
≤1 per month	0.710 [0.590, 0.860]	0.001	0.730 [0.580, 0.930]	0.009
1–3 per month	0.920 [0.690, 1.230]	0.566	1.020 [0.720, 1.430]	0.922
1–6 per week	0.780 [0.610, 1.000]	0.047	0.670 [0.490, 0.900]	0.009

Journal Pre-proof

Journal Pre-proof

Predictor	SRH, OR [95% CI]	*p*	IADL, OR [95% CI]	*p*
≥1 per day	0.590 [0.510, 0.700]	< 0.001	0.670 [0.560, 0.820]	< 0.001
Smoking status				
Never	ref.		ref.	
Former	1.520 [1.280, 1.820]	< 0.001	1.520 [1.240, 1.860]	< 0.001
Current	1.140 [0.990, 1.300]	0.067	1.120 [0.950, 1.310]	0.181
Chronic illness				
No	ref.		ref.	
Yes	3.690 [3.310, 4.100]	< 0.001	2.470 [2.180, 2.810]	< 0.001
Random effects				
Variance (intercept)	2.350 [2.210, 2.510]		2.820 [2.610, 3.040]	

Note. Number of children was mean-centered (*M* = 3.36, *SD* = 1.52). Odds ratios above 1 indicate poorer self-rated health or greater IADL limitation severity; IADL limitations were modeled on the original 0-to-5 scale. Observations: 23,893 for SRH and 24,943 for IADL limitations. CI = confidence interval; IADL = instrumental activities of daily living; ref. = reference category; SRH = self-rated health.

Declaration of_competing_interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Journal Pre-proof